

Assignment 2

Robotics in Medicine and Industry

April 4, 2025

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1 Executive Summary

This report outlines an automated system for the retrieval, processing, and storage of different types of sports car rims. The project is divided into two key tasks: image classification using deep learning, and the design of a fully automated industrial process involving rim retrieval, deburring, polishing, flipping, and organized storage.

In Task 1, a Convolutional Neural Network (CNN) was used to classify sports rims from a custom dataset of 150 images across 10 classes. Transfer learning was applied with five pre-trained architectures—ResNet18, InceptionV3, MobileNetV2, AlexNet, and VGG16—where ResNet18 proved to be the most effective. Additionally, a hyperparameter sensitivity analysis was conducted, providing valuable insights into each model’s adaptability. The classification data generated serves as critical input for the automation control logic in Task 2.

Task 2 introduces a fully autonomous system for rim handling and finishing, integrating advanced robotics and intelligent automation. Rim retrieval from an unorganized storage environment is achieved using drone-assisted image recognition and LiDAR-based depth mapping. A 6-DoF robotic arm mounted on a linear axis transports the selected rim to a deburring station, where the optimal program is automatically selected based on classification data. To ensure efficient polishing on both sides, a passive gravity-assisted flipping mechanism is incorporated, eliminating the need for actuators. MATLAB simulations validated its reliability.

The polishing station utilizes a second 6-DoF robot with an automatic tool exchange system, alternating between a microfiber cleaning head and a polishing pad. A force-torque sensor paired with an Active Contact Flange (ACF) ensures real-time force adjustment for consistent surface treatment. Microfiber pads, chosen for their flexibility and high-quality finish, operate within an optimized force and speed range. The process concludes with AMR-based transport to a structured vertical storage system, minimizing manual handling and improving scalability.

The proposed system offers a highly automated and flexible solution adaptable to various rim types while minimizing process bottlenecks through asynchronous deburring and polishing stages. A cost analysis estimates the total investment at \$96,200–\$196,900, justified by long-term savings in labor, defect reduction, and improved processing efficiency. The report also identifies limitations including surface finish compatibility and spatial constraints, and proposes future recommendations for integrating wax polishing and a vertically stacked flipping mechanism to enhance performance.

2 Task 1: Image Classification

Image classification is a key technology in both robotics and industrial applications, allowing machines to recognize and categorize objects within images. A widely used approach for image classification is Convolutional Neural Networks (CNNs). They are a type of deep learning model specifically designed for processing grid-like data, such as images. This section looks at training a CNN for wheel rim classification.

2.1 Dataset

The dataset collection process involved identifying ten classes of wheel rims, namely - Porsche Carrera, Audi A3, Volkswagen Golf MK8, BMW 530e, Benz E350, Tesla R588GB, Ferrari GTC4, Lamborghini Huracan Giano, Jaguar Vulcan, and Nissan Leaf. Fifteen images were downloaded for each class, resulting in a total of 150 images in the dataset. The images were sourced from various public image repositories on the internet, and undergo pre-processing to make them standard. The dataset was further split as 70% training data (105 images) and 30% testing data (45 images).

The applied transformations enhance dataset variability and model generalization. The image is resized to 224x224 or 299x299 pixels based on the model (e.g., VGG uses 224x224, Inception uses 299x299), followed by cropping if needed. Random transformations like horizontal flipping, rotation (up to 15 degrees), and affine adjustments (translation and scaling) introduce further variation. Color jitter adjusts brightness, contrast, and saturation to simulate different lighting conditions. Finally, the image is converted to a tensor and normalized using ImageNet's mean and standard deviation for model compatibility.



Figure 1: Overview of ten classes in the dataset

2.2 Model Selection

Initially, the team considered training a custom CNN from scratch. However, due to the limited size of the dataset, leveraging transfer learning with a pre-trained model would be a more effective approach. By fine-tuning only the final layers of a pre-trained model, we could take advantage of the rich feature representations learned from large-scale datasets, thereby improving performance while minimizing the risk of overfitting on the smaller dataset.

Five models were selected for training the classifier: ResNet18, InceptionV3, MobileNetV2,

AlexNet, and VGG16. These models were chosen due to their widespread use and proven success in image classification tasks. By evaluating these diverse architectures, a comparative analysis can be conducted to identify the best-performing model for this specific application.

2.3 Experiment

The goal of this experiment is to evaluate the performance of five different deep learning classifiers on a 150-image wheel rim dataset, with a focus on comparing their classification efficacy. To establish a fair baseline for comparison, we optimized the training parameters using the ResNet18 architecture, which has demonstrated robust performance across diverse computer vision tasks. The optimal hyperparameters identified (learning rate = 0.003, batch size = 8) were then applied uniformly across all five model architectures. This controlled approach allows us to isolate the impact of architectural differences on wheel rim classification performance while maintaining consistent training conditions.

2.4 Model Training

Models were initialized with ImageNet-pretrained weights to leverage learned features for faster convergence. Early convolutional layers were frozen to retain pre-learned features, while custom fully-connected layers replaced the original classification layers to tailor the model for wheel rim classification.

2.4.1 Training factors

1. **Learning Rate:** Followed a differential approach (0.0003 for pretrained layers, 0.003 for new classifier layers) to help balance fine-tuning pretrained layers and training the new classifier.
2. **Batch Size:** Set to 8 or 12 to balance model stability and training time; smaller batches help in generalization.
3. **Optimizer:** Utilized Adam to combine the advantages of both Adagrad and RMSprop, offering adaptive learning rates.
4. **Learning Rate Schedule:** Step-decay ($\gamma=0.5$) was implemented every 7 epochs, which reduces learning rate gradually to prevent overshooting at later stages.
5. **Cross-Entropy Loss Function:** It measures the difference between predicted class probabilities and true labels, guiding model updates during training.
6. **Dropout:** 0.5-0.6 rates between fully-connected layers – It randomly drops neurons during training to prevent overfitting.
7. **Batch Normalization:** Applied after ReLU activations – It normalizes inputs to each layer for stable and faster training.
8. **Label Smoothing:** Factor of 0.1 with cross-entropy loss – It reduces model confidence for class predictions, improving generalization.

2.4.2 Training Dynamics

Models were trained for 50 epochs with continuous tracking of weights yielding the lowest training loss, effectively implementing early stopping. This ensures the model doesn't overfit by halting training when performance plateaus.

2.5 Model Validation

The testing dataset was employed to validate each of the five models. They were assessed on the following metrics which look at accuracy and performance:

2.5.1 Training Metrics

1. Training Loss & Accuracy per Epoch: Tracks loss and accuracy across epochs.
2. Training Time: Measures total training duration
3. Best Model Selection: Saves the model with the lowest training loss.
4. Training Curves (Loss & Accuracy over Epochs): Plots training loss and accuracy trends

2.5.2 Testing Metrics

1. Final Test Accuracy & Loss: Evaluates overall performance on test dataset.
2. Class-wise Accuracy: Reports accuracy for each class
3. Inference Time: Measures average processing time per image
4. Confusion Matrix: Visualizes model predictions versus actual labels
5. High Confidence Predictions: Computes accuracy for predictions with confidence greater than 0.7.

2.6 Classification Results

The following results were determined from the five classifiers.

2.6.1 ResNet18

Metric	Value
Accuracy	0.8000
Cross Entropy Loss	1.0326
Lowest Training Loss	0.6554
Inference Time	0.0042
Total Training Time	1m 4s
Confidence Prediction Accuracy	0.9231
Confidence Predictions Made	26/50

Table 1: Model Performance Metrics for ResNet18

Class	Score	Class	Score
Audi	40.00%	Lamborghini	100.00%
BMW	100.00%	Nissan	80.00%
Benz	100.00%	Porsche	80.00%
Ferrari	80.00%	Tesla	80.00%
Jaguar	100.00%	Volkswagen	40.00%

Table 2: Class-wise Accuracy for ResNet18

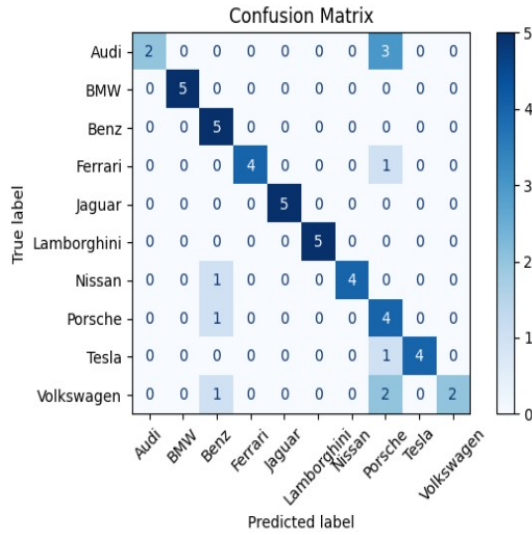


Figure 2: Confusion Matrix for ResNet18

2.6.2 InceptionV3

Metric	Value
Accuracy	0.7000
Cross Entropy Loss	1.4205
Lowest Training Loss	1.1619
Inference Time	0.0151
Total Training Time	1m 42s
Confidence Prediction Accuracy	1.0000
Confidence Predictions Made	3/50

Table 3: Model Performance Metrics for InceptionV3

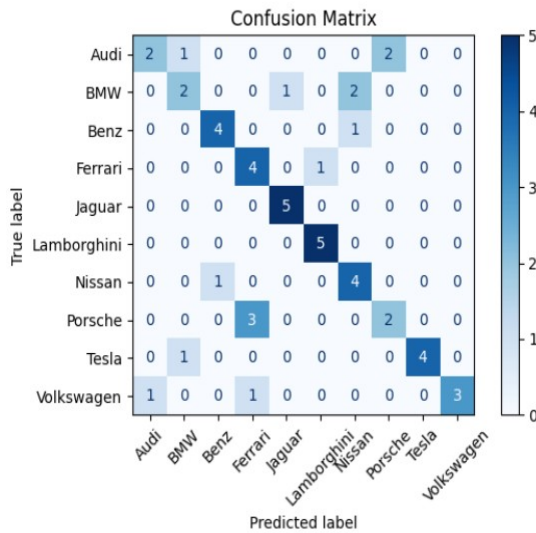


Figure 4: Confusion Matrix for InceptionV3

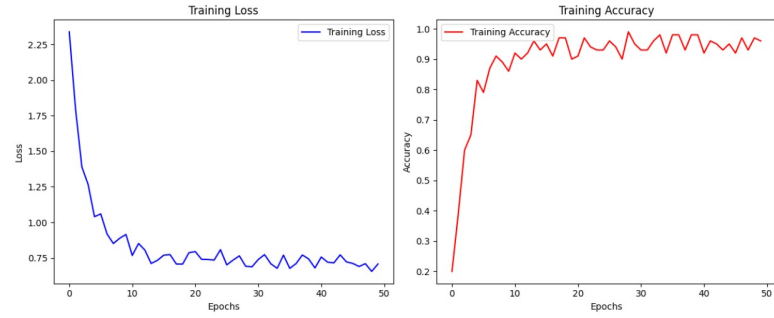


Figure 3: Accuracy and Loss curves for ResNet18

Class	Score	Class	Score
Audi	40.00%	Lamborghini	100.00%
BMW	40.00%	Nissan	80.00%
Benz	80.00%	Porsche	40.00%
Ferrari	80.00%	Tesla	80.00%
Jaguar	100.00%	Volkswagen	60.00%

Table 4: Class-wise Accuracy for InceptionV3

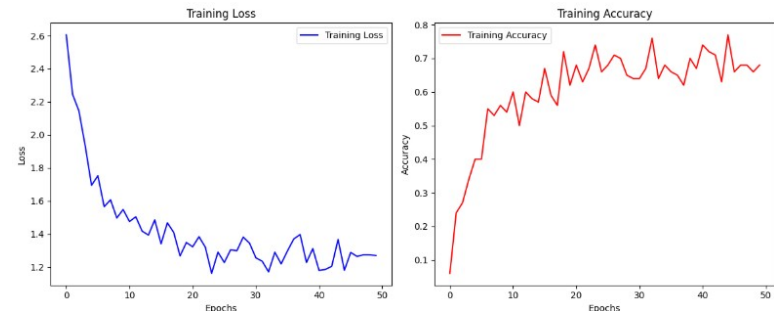


Figure 5: Accuracy and Loss curves for InceptionV3

2.6.3 MobileNetV2

Metric	Value
Accuracy	0.7600
Cross Entropy Loss	1.1066
Lowest Training Loss	0.8315
Inference Time	0.0092
Total Training Time	0m 58s
Confidence Prediction Accuracy	1.0000
Confidence Predictions Made	19/50

Table 5: Model Performance Metrics for MobileNetV2

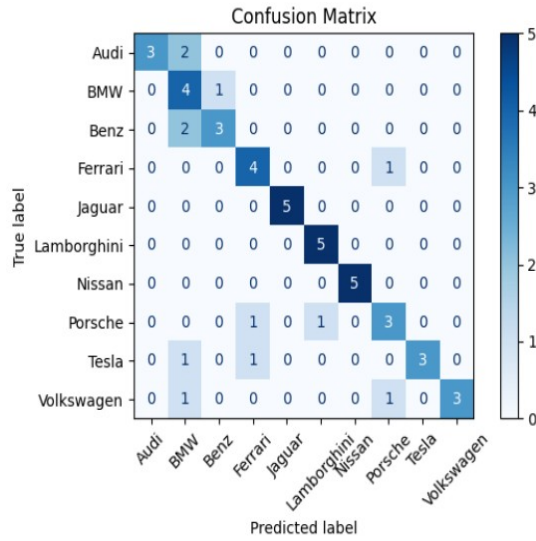


Figure 6: Confusion Matrix for MobileNetV2

2.6.4 AlexNet

Metric	Value
Accuracy	0.7800
Cross Entropy Loss	1.0415
Lowest Training Loss	0.6770
Inference Time	0.0013
Total Training Time	1m 7s
Confidence Prediction Accuracy	0.9231
Confidence Predictions Made	26/50

Table 7: Model Performance Metrics for AlexNet

Class	Score	Class	Score
Audi	60.00%	Lamborghini	100.00%
BMW	80.00%	Nissan	100.00%
Benz	60.00%	Porsche	60.00%
Ferrari	80.00%	Tesla	60.00%
Jaguar	100.00%	Volkswagen	60.00%

Table 6: Class-wise Accuracy for MobileNetV2

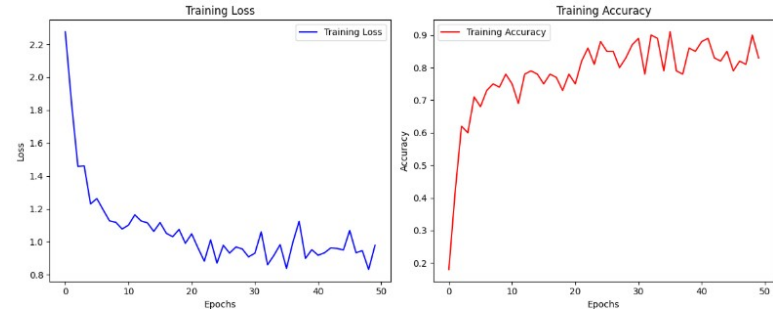


Figure 7: Accuracy and Loss curves for MobileNetV2

Class	Score	Class	Score
Audi	40.00%	Lamborghini	100.00%
BMW	100.00%	Nissan	100.00%
Benz	60.00%	Porsche	80.00%
Ferrari	80.00%	Tesla	100.00%
Jaguar	80.00%	Volkswagen	40.00%

Table 8: Class-wise Accuracy for AlexNet

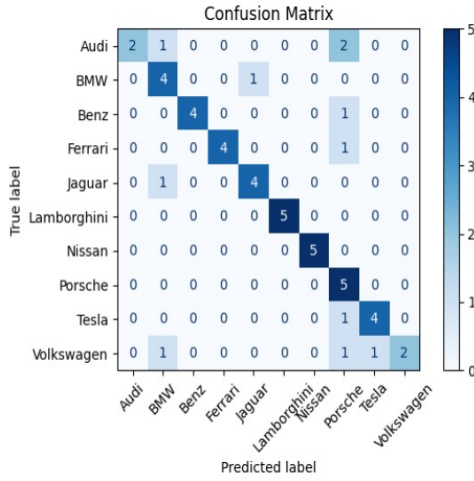


Figure 8: Confusion Matrix for AlexNet

2.6.5 VGG16

Metric	Value
Accuracy	0.6600
Cross Entropy Loss	1.3859
Lowest Training Loss	1.0610
Inference Time	0.0020
Total Training Time	1m 47s
Confidence Prediction Accuracy	0.9231
Confidence Predictions Made	26/50

Table 9: Model Performance Metrics for VGG16

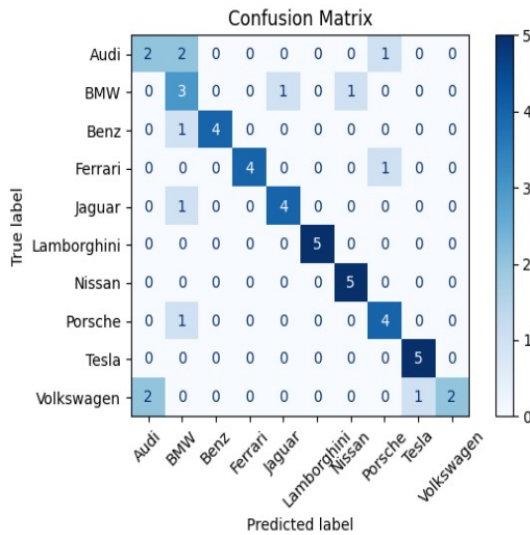


Figure 10: Confusion Matrix for VGG16

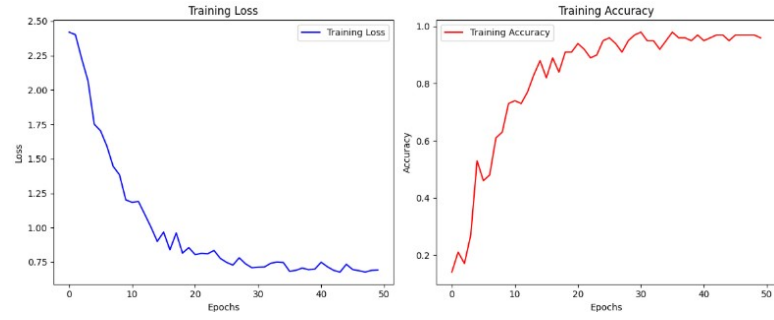


Figure 9: Accuracy and Loss curves for AlexNet

Class	Score	Class	Score
Audi	40.00%	Lamborghini	80.00%
BMW	40.00%	Nissan	100.00%
Benz	40.00%	Porsche	20.00%
Ferrari	100.00%	Tesla	80.00%
Jaguar	100.00%	Volkswagen	20.00%

Table 10: Class-wise Accuracy for VGG16

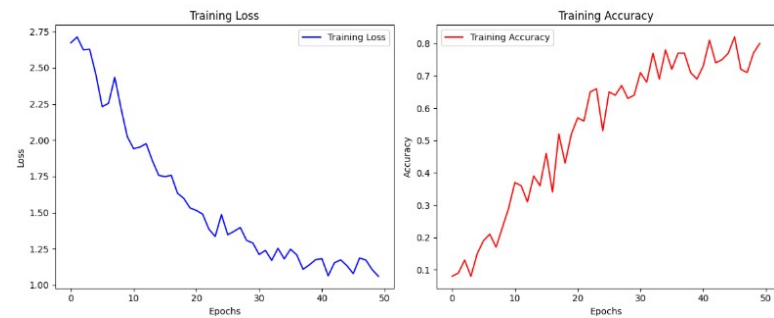


Figure 11: Accuracy and Loss curves for VGG16

2.7 Analysis of Results

The experiments conducted with five different CNN architectures (ResNet18, InceptionV3, MobileNetV2, AlexNet, and VGG16) for wheel rim classification provide valuable insights into their relative strengths and weaknesses.

2.7.1 Overall Accuracy Comparison

Among the tested models, ResNet18 achieved the highest overall accuracy at 80.0%, closely followed by AlexNet at 78.0% and MobileNetV2 at 76.0%. InceptionV3 and VGG16 showed comparatively lower performance at 70.0% and 66.0% respectively.

Model	Test Accuracy	Test Loss	Inference Time (s)	Training Time
ResNet18	80.0%	1.0326	0.0042	1m 4s
AlexNet	78.0%	1.0415	0.0013	1m 7s
MobileNetV2	76.0%	1.1066	0.0092	2m 58s
InceptionV3	70.0%	1.4205	0.0151	1m 42s
VGG16	66.0%	1.3859	0.0020	1m 47s

Table 11: Model Performance Metrics

The results suggest that the complex architectures of InceptionV3 and VGG16 may not necessarily translate to better performance for this specific task. Despite being deeper networks with more parameters, they were outperformed by the relatively simpler ResNet18 and AlexNet architectures.

2.7.2 Efficiency Analysis

AlexNet demonstrated the fastest inference time (0.0013s), making it approximately 11.6 times faster than InceptionV3 (0.0151s). MobileNetV2, despite being designed for mobile applications with efficiency in mind, showed moderate inference speed (0.0092s) but had the shortest training time (58s). This makes AlexNet particularly suitable for real-time applications where processing speed is crucial. The efficiency-accuracy tradeoff is apparent when comparing MobileNetV2 and ResNet18. While ResNet18 achieved better accuracy, MobileNetV2 trained faster and could be preferable for resource-constrained environments.

2.7.3 Class-wise Performance Analysis

1. **Consistently Well-Classified Brands** - Certain car brands were consistently well-classified across all models:
 - **Lamborghini:** Achieved 100% classification accuracy across all tested models, indicating highly distinctive rim designs.
 - **Jaguar:** Performed excellently with 100% accuracy on ResNet18, InceptionV3, and MobileNetV2, and 80% on AlexNet.
 - **Ferrari:** Maintained a consistent 80% accuracy across all models.
2. **Problematic Classifications** - Some brands proved challenging for most models:
 - **Audi:** Consistently among the lowest performing classes, with accuracy ranging from 40-60% across different models.
 - **Volkswagen:** Similarly difficult to classify correctly, with accuracy ranging from 40-60%.

- **Porsche:** Showed variable performance across models (40-80% accuracy).

The confusion between Audi and Volkswagen may be attributed to their shared corporate design language as both brands belong to the Volkswagen Group. This suggests that wheel rims within the same automotive group might share design characteristics that make classification more challenging.

2.7.4 Confidence Analysis

The high-confidence prediction accuracy provides valuable insights into model reliability. ResNet18, AlexNet, and VGG16 all achieved 92.31% accuracy for high-confidence predictions, with 26 out of 50 test samples exceeding the confidence threshold. MobileNetV2 and InceptionV3 achieved perfect accuracy (100%) for high-confidence predictions, though InceptionV3 was notably cautious with only 3 high-confidence predictions.

This suggests that while InceptionV3's overall accuracy was lower, it exhibited excellent calibration – when it was confident in its prediction, it was invariably correct. This characteristic could be valuable in applications where false positives are particularly costly.

2.8 Hyperparameter Sensitivity

To investigate hyperparameter sensitivity across architectures, we systematically modified two hyperparameters—increasing batch size to 12 and learning rate to 0.005—and measured the resulting performance changes across all models. This analysis serves multiple purposes: it reveals which models might benefit most from custom optimization, and provides insights into the robustness of different architectures when deployed in real-world scenarios where optimal tuning might not always be possible.

Model	Original Accuracy	Modified Accuracy	Change
ResNet18	80.0%	82.0%	+2.0%
AlexNet	78.0%	82.0%	+4.0%
MobileNetV2	76.0%	72.0%	-4.0%
InceptionV3	70.0%	68.0%	-2.0%
VGG16	66.0%	60.0%	-6.0%

Table 12: Comparison of Original and Modified Accuracies for Different Models

The analysis shows ResNet18 and AlexNet benefit from more aggressive tuning, with improvements of 2% and 4% respectively, while VGG16, InceptionV3 and MobileNetV2 show declines due to their architectural limitations. This hyperparameter sensitivity analysis builds on the earlier performance comparisons by identifying the best-performing models under optimal conditions and highlighting those that are most adaptable to changes in parameters—a key factor for real-world applications where conditions and constraints evolve.

2.9 Limitations and Future Work

The relatively small number of samples per class (indicated by only 5 test samples per class) may limit the generalizability of these findings. Also, rim classification performance may vary based on image quality, lighting conditions, and camera viewpoint.

Future work could address these limitations by expanding the dataset with more examples per class and potentially more classes. It could also implementing model ensemble techniques to leverage the strengths of multiple architectures, and better capture the fine-grained details that distinguish similar wheel rim designs.

3 Task 2: Automation Process Design

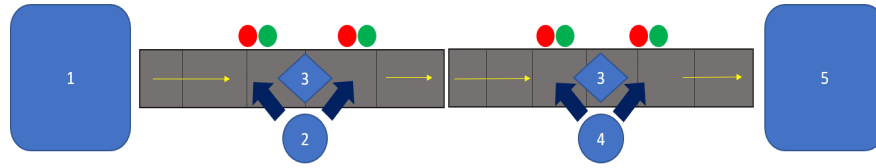


Figure 12: Floor plan of autonomous process design: (1) Disorganized Storage (2) Deburring Station (3) Flipping Mechanism (4) Polishing Station (5) Organized Storage

3.1 Rim Retrieval From Disorganized Storage

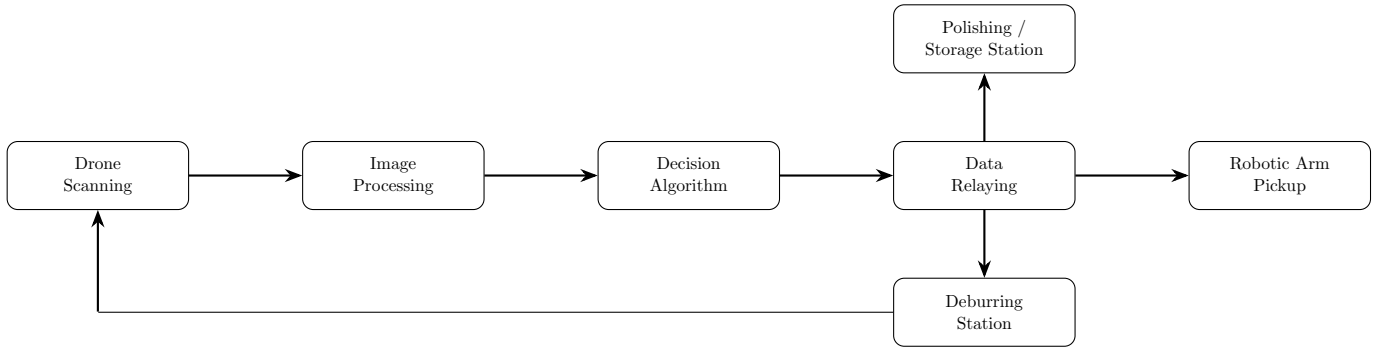


Figure 13: High level rim retrieval flowchart

3.1.1 Drone Scanning

The rim retrieval process begins with a swarm of autonomous drones flying over the disorganized storage area in a coordinated grid-based scan pattern. Each drone is equipped with an onboard high-resolution RGB camera that captures detailed images from multiple angles, enabling the system to clearly distinguish between different rim patterns, diameters, and orientations. Operating in parallel, the drones rapidly scan the entire storage space, significantly reducing detection time. To complement the visual data, the drones also utilise integrated LiDAR (Light Detection and Ranging) depth sensors, which emit laser pulses and measure the time it takes for the reflections to return. This process generates a comprehensive 3D depth map of the environment, allowing the system to determine the precise spatial coordinates and orientation of each rim—whether lying flat, leaning, or standing upright.

3.1.2 Image Processing & Decision Algorithm

Once the drones capture visual and depth data, onboard image processing begins immediately. Each drone is equipped with an embedded CNN trained to classify the 10 different types of sports rims. The CNN processes the high-resolution RGB images in real time, identifying the rim type based on pattern, diameter, and width. Simultaneously, the drone's software analyses the LiDAR-generated depth map to determine the rim's orientation and exact location in 3D space.

Following this, the drone runs a decision-making algorithm that evaluates the detected rims based on several key criteria:

1. **Shortest travel distance** from the current position of the robotic arm pickup to minimise retrieval time.

2. **Surface accessibility**, favouring rims that are fully visible and accessible, with minimal obstruction or blockage from surrounding rims or objects.
3. **Rim clustering**, avoiding selections that are tightly surrounded by other rims, which may increase the risk of collision during retrieval.
4. **CNN classification confidence**, giving preference to rims with a high confidence score from the classification model to reduce misidentification risk; lower confidence may result from poor lighting, partial blocking, or unfavourable viewing angles.

The algorithm assigns a priority score to each rim based on these factors and selects the single top-ranked rim for retrieval.

3.1.3 Data Relaying & Robotic Arm Pickup

Once the optimal rim is selected by the decision-making algorithm, the classification data, including rim type and required processing parameters, is transmitted to the relevant workstations—deburring, polishing, and final storage. This early communication allows each station to preload the necessary programme settings corresponding to the specific rim type before the physical rim even arrives. By doing so, the system effectively minimises downtime and reduces potential bottlenecks at each station. After the deburring station completes processing a rim, it sends a confirmation signal back to the drone system, triggering a new scanning cycle. The drone swarm then rescans the area, reevaluates the remaining rims using the same prioritisation criteria, and selects the next optimal rim for retrieval. This creates a continuous, closed-loop feedback system that keeps the workflow efficient and responsive.

Simultaneously, detailed data including the classified rim type, exact 3D coordinates, and orientation is also relayed to the robotic arm mounted on a motorised external linear axis to extend its operational workspace as seen below in Fig. 14. Based on the orientation data, the robotic arm dynamically selects the appropriate gripping method, as shown in Table 13, and plans the pickup accordingly—using a 2-finger gripper for upright rims, a 3-finger gripper for flat-lying rims, and a hybrid strategy for leaning or skewed rims. Once the rim is successfully grasped, the robot begins moving along its external linear track and transports the rim directly to the deburring station.

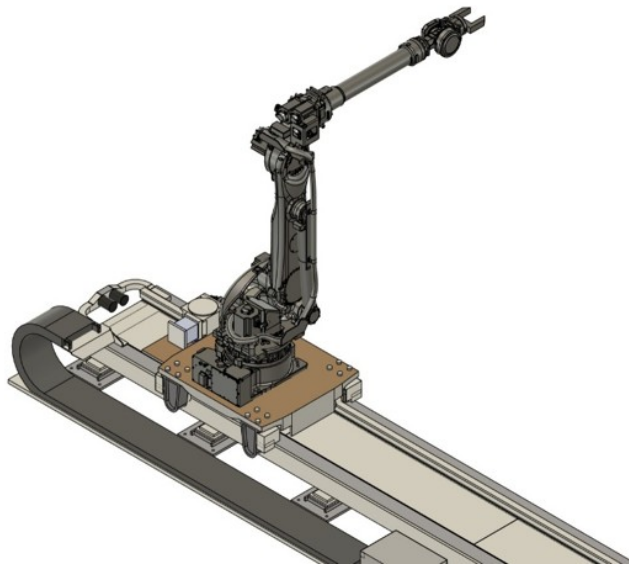


Figure 14: Robotic arm mounted on a linear external axis for extended workspace

Orientation	Gripping Method	Action Description
Flat (face-up/down)	3-finger mechanical gripper	The gripper engages three equidistant points on the outer arc of the rim, ensuring stable lifting from a flat position.
Upright (rim standing vertically)	2-finger mechanical gripper	The gripper clamps onto two opposite points of the outer arc of the rim, allowing secure side pickup and vertical lift.
Leaning/skewed	Hybrid approach	The arm first aligns to a stable approach angle using visual and orientation data, then selects between the 2- or 3-finger gripper based on contact feasibility.

Table 13: Gripping method based on rim orientation

Component	Estimated Cost (\$)
4x Drones with LiDAR Scanners	10,000–16,000
Embedded CNN Processor (Edge AI Module)	800–1,500
Retrieval Control System	3,000–6,000
6-DoF Robotic Arm (for pickup)	12,000–25,000
3-Finger Adaptive Gripper	2,000–4,000
Motorised External Linear Axis	5,000–10,000

Table 14: Estimated cost breakdown for components used in the rim retrieval system.

3.2 Deburring Station

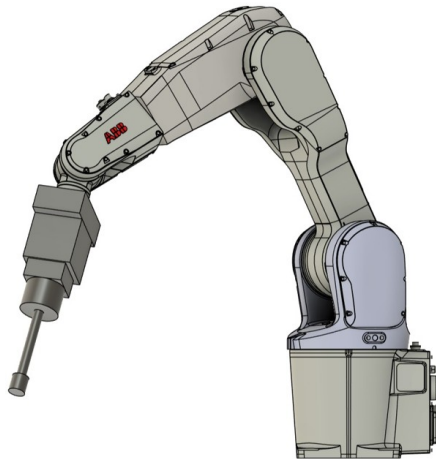


Figure 15: Deburring tool CAD model

The deburring process requires flexibility and precision, especially when handling rims with complex and varied geometries. To meet these demands, a 6-degree-of-freedom (6-DoF) robotic arm, shown above in Figure 15, is employed at the deburring station. This configuration allows the robot to position and orient the tool in three-dimensional space, enabling it to contour around the inner and outer edges of the rim with ease. The increased articulation is particularly important for reaching recessed areas, following curved profiles, and transitioning smoothly between surfaces—tasks that simpler arms or fixed tools would struggle with. To ensure consistent and accurate deburring, the rim must remain completely stationary throughout the process. Therefore, the station is equipped with adjustable mechanical clamps that

securely hold the rim in place, preventing any movement or vibration that could compromise the finish or tool alignment.

Maintaining consistent pressure between the deburring tool and the rim is critical for both surface quality and process safety. To achieve this, the robotic arm is equipped with a force-torque sensor mounted near the end-effector. This sensor continuously monitors the interaction forces during the deburring process and adjusts the arm's movement in real time to prevent excessive pressure or chatter. This dynamic control helps to ensure that material is removed evenly without damaging the rim, even when surface irregularities or inconsistencies are present. It also allows the robot to adapt to slight variations in rim shape across different batches.

Examples of commonly used sensors in such applications include the ATI Industrial Automation Axia80 [1] and the OnRobot HEX-E [2]. The ATI Axia80 offers high-resolution six-axis force and torque measurement and is widely used in deburring, grinding, and polishing tasks. Similarly, the HEX-E sensor from OnRobot provides precise force control and is designed specifically for tasks requiring constant contact, like surface finishing. These sensors integrate easily with most industrial robot arms and support advanced control strategies such as impedance or admittance control to maintain a safe and adaptive process.

Each sports rim requires a tailored deburring approach depending on its material composition, thickness, diameter, and surface complexity. To ensure this, the system uses a fully automated process in which the rim is first classified by the drone-based CNN model before physical retrieval as mentioned before. Once the drone has identified the rim type, this information is relayed in advance to the deburring station. This early communication allows the appropriate deburring programme—consisting of pre-set toolpaths, rotational speeds (RPM), contact pressure, and tool engagement strategies—to be automatically loaded into the robotic control system ahead of time. By the time the robotic arm delivers the rim to the station, the required parameters are already in place, effectively reducing downtime and avoiding bottlenecks in the workflow.

The rotary deburring tool at the station features adjustable RPM settings to suit the specific characteristics of each classified sports rim. For example, a lightweight forged rim with narrow spokes—typical of performance vehicles—requires a lower rotational speed and gentler pressure to avoid damaging the surface or fine details. On the other hand, a wider, solid-faced sports rim designed for endurance or track use may tolerate higher RPMs and more aggressive material removal. The flexibility to tailor RPM and force settings dynamically ensures that every rim is processed efficiently and to the correct quality standard without requiring manual adjustments between units.

Table 15 outlines the recommended deburring RPM ranges for various types of sports rims, illustrating how tool speed is tailored based on design features such as spoke density, rim geometry, and surface complexity. The rotational speed of the deburring tool is determined based on surface speed requirements expressed in Surface Feet Per Minute (SFPM). According to industry guidelines, an effective range for deburring operations lies between 5000–6500 SFPM [3]. This range provides sufficient energy for material removal without causing thermal damage or surface deformation, particularly important when working with lightweight aluminium sports rims.

To convert surface speed into RPM, the following equation is used:

$$\text{Rotational Speed, RPM} = \frac{\text{SFPM} \times 12}{\pi \times \text{Diameter (inches)}} \quad (1)$$

For example, with a 6-inch deburring wheel and a target surface speed of 5000 SFPM, the

resulting RPM is approximately 3183. Consequently, the optimal RPM varies depending on the tool diameter and must be adjusted accordingly for different rims and tool setups. This approach ensures consistent surface finish and efficient material removal tailored to each rim design.

Rim Type	Vehicle Example	Design Features	Recommended RPM
Lightweight Forged Rim	Porsche 911 Turbo	Thin spokes, low mass, detailed finish	800–1200 RPM
Multi-Spoke Racing Rim	BMW M4 Competition	High spoke count, narrow spoke gaps	1000–1400 RPM
Deep-Dish Performance Rim	Nissan GT-R	Wide lip, thick edge geometry	1400–1800 RPM
Solid-Face Track Rim	Audi RS3 LMS	Minimal openings, dense surface area	1600–2000 RPM
Aero-Optimised Sports Rim	Tesla Model 3 Performance	Curved surfaces, aerodynamic inserts	1200–1600 RPM

Table 15: Recommended deburring RPM ranges for different types of sports rims

As the rim undergoes deburring, fine metal particles and dust are generated, which can accumulate on the work surface and potentially interfere with other parts of the automation process. To manage this, the deburring station is equipped with an integrated debris extraction system. A vacuum hood is positioned close to the working area to capture loose debris as it is produced, maintaining a clean environment around the rim. This not only improves visibility for any inline inspection systems but also reduces wear on mechanical components and ensures compliance with safety and cleanliness standards.

Component	Estimated Cost (\$)
6-DoF Industrial Robot Arm	12,000–25,000
Force-Torque Sensor	3,000–6,000
Deburring End-Effector	1,000–3,000
Debris Management System (Vacuum)	800–2,000
Mechanical Clamps for Rim Holding	500–1,200
Deburring Control System	3,000–7,000

Table 16: Estimated cost breakdown for key components of the deburring station.

3.3 Flipping Mechanism

In order to debur and polish both sides of the rims, a flipping mechanism must be implemented. This flipping mechanism is a gravity-assisted system that enables the rims to be flipped in preparation for deburring and polishing without the need for actuators and robotic manipulators. This mechanism is integrated directly into the conveyor system to streamline the process and minimize handling time and cost.

The mechanism relies on a gradually curving conveyor segment that tilts the rim past its critical angle—approximately $100 - 110^\circ$. Once this threshold is passed, the rim's centre of gravity shifts beyond its base of support, causing it to flip over under the influence of gravity and momentum. This effect is predictable and consistent across different rim types, as gravitational torque increases with tilt until the object becomes unstable and tips over. The torque applied is given by:

$$\tau = mgr \sin \theta \quad (2)$$

where:

- m = mass of rim (kg)
- g = gravitational acceleration (ms^{-2})
- r = rim's radius (m)
- θ = tilt angle of conveyor ($^\circ$)

The torque applied reaches a maximum once the rims are tilted 90° , once they get tilted passed that, gravity will force the rims to flip, as previously explained. Fig.16 is an image of the structure, containing snap shots of the 'rim flipping'.

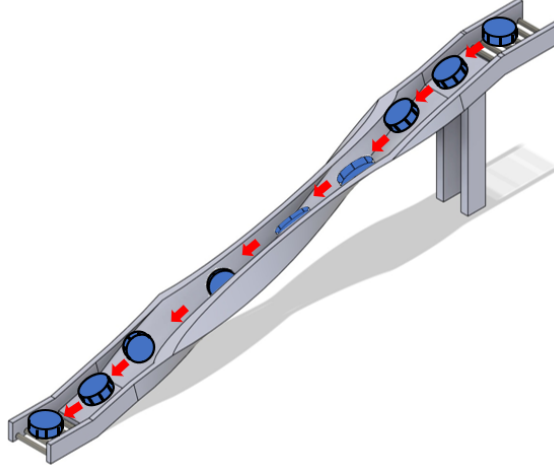


Figure 16: The flipping structure

For this tilting to happen gradually, and successfully, we need to calculate the length of the curved conveyor which is responsible for the tilting. It must happen gradually to allow the rims to tilt, however, without taking a very large length of the conveyor belt, to allow the deburring and polishing robots to be able to reach to the rims before and after tilting, as seen in Fig. 12. To calculate this length, Fig.17 below confirms that this flipping will occur reliably across some tested rim sizes and weights, with sufficient torque generated to complete the flip without excessive stress when the curved length of the conveyor is about $2m$ long, while the angular increase of the curved conveyor length increases linearly ($55^\circ/m$).

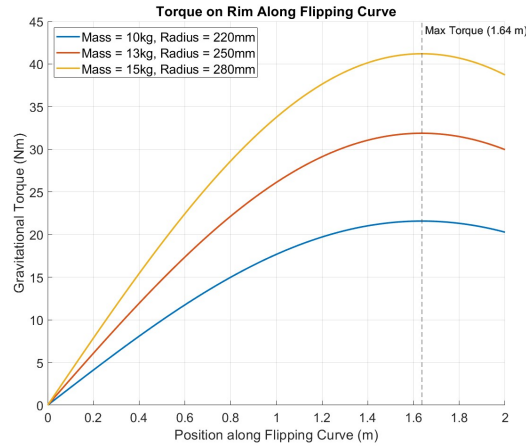


Figure 17: MATLAB plot of the torque applied on different types of rims

To enable the rim to successfully flip, the curved section of the conveyor must achieve a total angular tilt of at least 90° . Assuming a constant angular gradient of $55^\circ/m$, this critical angle is reached at approximately 1.64 meters along the curve. To ensure reliable flipping across all rim types and account for any variations in rim geometry or frictional effects, a conservative total curve length of 2 meters has been selected.

To mitigate the risk of damage caused by impact upon flipping, the receiving side of the conveyor is also designed with a curved, downward-facing profile. This reduces the rim's kinetic energy during descent by gradually decelerating it, thus minimizing mechanical stress. Importantly, the gap between the outbound and inbound conveyor curves is wider than the width of a rim. This prevents scenarios where the rim might land upright — balanced with its cylindrical edge — which would inhibit flipping and proper alignment. The additional spacing ensures the rim fully inverts and settles safely onto the second conveyor segment.

Since the rim may not remain centrally aligned on the conveyor belt immediately after flipping, a short funnel-shaped section is incorporated downstream of the flipping mechanism. This tapered design gradually guides and re-centers the rim along the conveyor's central axis, ensuring accurate positioning. As a result, the rim arrives correctly aligned at the deburring and polishing stations, where precise placement is essential for effective and consistent surface treatment.

To prevent damage during flipping, the conveyor and contact surfaces along the curve are constructed from polyurethane (PU)-coated stainless steel. This combination offers high durability and stiffness to support rim weights, elasticity to dampen impact forces, and surface compliance that avoids rim scratches or dents. Polyurethane-coated belts are known for their excellent wear resistance, flexibility, and resistance to oil and grease, making them suitable for handling metal parts. Additionally, PU coatings provide a smooth surface that minimizes friction and wear on conveyed materials, further protecting them from scratches or damage [4].

For this application, a flat belt conveyor is selected due to its suitability for transporting irregularly shaped and heavy items like rims, providing an even surface that ensures stable support during movement, reducing the risk of tipping or misalignment [5].

To accurately detect the position of the rim as it arrives at the deburring and polishing stations, a photoelectric sensor system is integrated along the conveyor belt. This system consists of a laser emitter positioned on one side of the conveyor and a corresponding photodetector receiver on the opposite side, aligned at a fixed height relative to the rim's center.

As the rim travels along the conveyor, it eventually interrupts the laser beam. This obstruction causes the photodetector to cease receiving the signal, triggering a logical condition in the control system. This interruption is interpreted as the presence of a rim at the designated processing position.

The information regarding the type of rim, including its geometric characteristics, is transmitted from the drone-based identification system to the conveyor belt's control unit. With knowledge of the rim's radius, the control logic adjusts the stopping position dynamically. Specifically, when the rim interrupts the through-beam photoelectric sensor, the system introduces a calculated delay before halting the conveyor. This delay corresponds precisely to the rim's radius, allowing the rim to continue moving such that its geometric center aligns exactly with the center of the deburring and polishing sites. This adaptive approach ensures accurate localization of rims of varying sizes and enhances process stability by guaranteeing correct positioning during surface treatments.

Considering the Robotic arm transporting the rims from the disorganized storage to the conveyor belt may take longer to pick up different rims, the distance between each rim on

the conveyor belt may not be constant. Therefore, once the conveyor belt stops to allow the polishing processes, deburring may not be possible to occur in parallel to this. To address this, the system incorporates two separate conveyor segments—one dedicated to the deburring station and the other to polishing. This decoupling allows the polishing conveyor to halt independently for precise surface treatment while the deburring conveyor can continue operating asynchronously, thereby maximizing process efficiency and throughput despite the variability in rim handling times.

3.4 Polishing

Most cars use aluminium alloy rims. Aluminum alloy is light and strong, thus, reducing fuel consumption whilst improving performance [6]. The alloying element is typically nickel, magnesium or a combination of both. Compared to steel rims, aluminium alloy rims offer lighter weight, thus improving acceleration and stopping of the car. Additionally, it offers less strain on suspension components of the car and is a better dissipator of heat from braking components [6]. However, these rims may get dirty due to dust and oxidise, therefore, losing their original finish (shine) over time. Therefore, by regularly polishing them allows them to maintain their original shiny finish. Polishing removes imperfections, scratches and oxidation from the rims by using abrasive compounds to smoothen and restore the shine of the rim's surface [7]. However, before polishing, the rims must be thoroughly washed to remove road dirt and debris.

To polish complex aluminium rims automatically, a selection of criteria must be followed to do so efficiently and successfully: flexibility, force control, surface finish quality, tool integration, automation compatibility, reliability and scalability. Regarding flexibility, reliability and scalability, a 6 degree of freedom (DoF) robot will be used. 6-DoF offers full spatial manipulation across all XYZ planes, allowing the robot to orient the rotary loose microfiber around complex rim geometries and, during polishing, track precise contours across both flat and curved surfaces of the rim, adjusting approach angles dynamically. The robot can perform this process without the need of image processing, however, solely the confirmation of the control unit of the conveyor belt that the rim has been successfully transported to the polishing site. All the classification data of the given rims were sent to the polishing robot by the drone, data relaying the information. Additionally, the robot must be able to reach and have access to both the inner and outer surfaces of rims without the need of repositioning the rim or the robot base, therefore, the robotic arm must have a suitable length of about $2 - 2.5m$.

The pre-polishing process is very important as it will affect the quality and efficiency of the polishing process. Assuming the pre-polishing process was not followed, any debris or road dust that gets trapped under the polishing pad would cause scratches and prevent the even application of the polishing compounds, reducing their effectiveness. Therefore, to clean the rim before polishing, a spring loaded base plate with a microfiber towel wrapped around it will be the second tool.

The pre-polishing process ensures the final surface finish quality of rims. Before any abrasive polishing compound is applied, the rim must be thoroughly cleaned to brake dust and particulate debris that may otherwise scratch or damage the surface during polishing. The microfiber cloth, made of fine synthetic fibers (typically 80% polyester and 20% polyamide), is known for its excellent ability to lift and trap dirt without scratching the surface [8]. The spring mechanism in the base plate allows the cleaning head to maintain consistent contact pressure against the curved surface of the rim, accommodating variations in geometry and surface height. Other materials that can be used are wool, felt and terry cloth. These materials are not comparable to microfiber (soft, fine synthetic fibers) as they are not reliable

to cover curved rim surfaces and may cause heat build up. The tool will rotate at about 1000 – 1300 *RPM* [9] to keep the heat buildup low, avoid excessive friction which would wear out the pad, and be better controlled when paired with the force-limited system.

Once the cleaning stage is complete, the robotic arm must automatically exchange the cleaning tool for the polishing tool to process with the next step. This is achieved using an automatic tool changer (ATC), such as those developed by ATI Industrial Automation [10]. The tool changer is mounted at the end of the robotic arm and includes a locking mechanism (spindle-lock) that enables secure attachment and detachment of both end-effectors [10]. When the cleaning operation finishes, the robot returns to a docking station where the microfiber cleaning tool is deposited and the polishing tool—which includes a rotary-driven microfiber polishing pad and an integrated fluid delivery nozzle—is picked up.

The table below depicts the key safety and performance features for the robotic polishing tool.

Key safety & performance features	Function	Benefit to Robotic Polishing Tool
Soft Start	Gradual increase of motor speed at startup	Prevents torque spikes that could cause rim damage or mechanical stress on the robot's end-effector
Constant Drive Control	Maintains steady RPM under varying loads	Ensures uniform surface finish despite rim contour changes or tool pressure variations.
Temperature Monitoring	Monitors internal motor/spindle temperature	Prevents overheating of tool and rim; protects them by reducing RPM and force once a temperature threshold is reached
Overload Protection	Shuts down or limits tool if torque/load exceeds safe limit	Protects the tool and rim in case of misalignment or excessive force during polishing
Restart Protection	Prevents tool from restarting after power failure without manual reset	Adds safety during automated operation, especially during power glitches or restarts
Spindle-lock	Locks tool spindle for safe and fast tool exchange	Enables secure automatic tool changing between cleaning and polishing heads
Rotary Motion	Enables spinning of polishing pad	Distributes polishing compound and creates even surface finish across rim geometry

Table 17: Key safety & performance features of polishing pad

To maintain a constant and safe force exerted on the rims a force-torque (F/T) sensor will be implemented on the robot [11]. This sensor will be placed between the polishing tool and the robotic arm's wrist (6th joint). This way, the sensor moves with the tool, allowing it to measure the forces directly acting between the robot and the rim surface, providing real-time feedback on how much force and rotating action is being applied during surface contact. The force applied on the rim can be divided into two components, tangential force (F_T) and normal force (F_N). Fig. 18a depicts the polishing force acting on the rim and Fig. 18b how each force component is controlled.

The wheel speed is used to control F_T as seen in the tree diagram in Fig. 18b. F_N is controlled by adjusting the position of the tool to the rim, by moving it further in or out the rim surface. Once the F/T sensor measures in real-time the force exerted on the rim, this information is sent to the force feedback control system. A proposed control system could be one manufactured by Ferrobotics, active contact flange (ACF) model ACF110-01

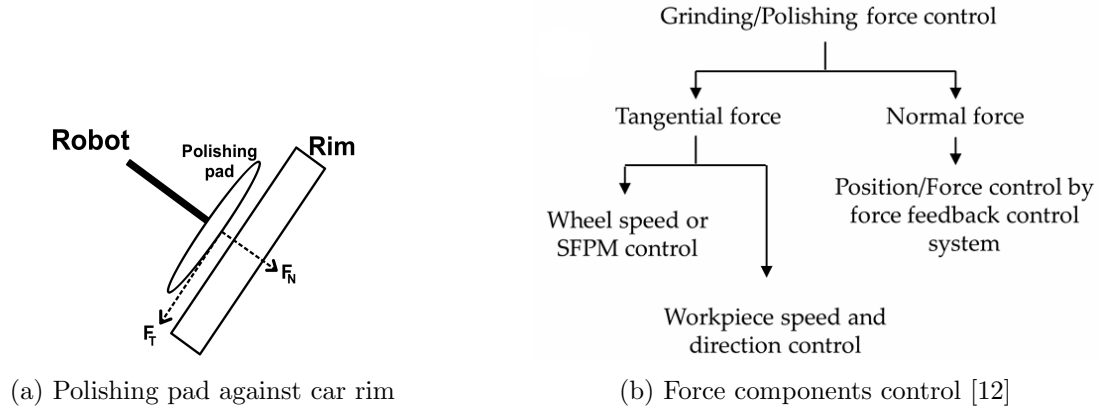


Figure 18: Polishing force

[12]. The control system compares the set threshold force with the measured force. The error signal is used to adjust the tool's position dynamically with a pneumatic actuator, keeping the polishing force constant even with changing surface geometry or robot arm orientation. The block control diagram is depicted in Fig. 19.

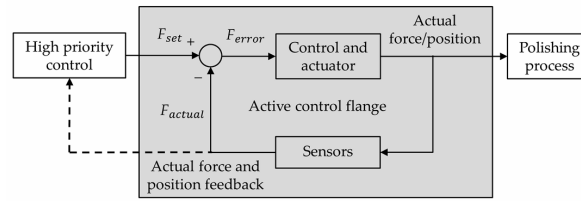


Figure 19: ACF control block diagram [12]

The tangential force on the other hand is controlled by adjusting the tool's rotational speed, which is done by adjusting the RPM of the end-effector.

Finally, to summarise the polishing process, the flowchart in Fig. 20, depicts the steps that need to be followed in order to polish the rim efficiently and successfully.

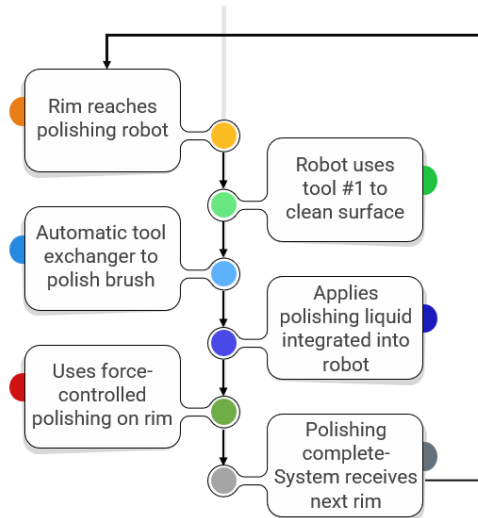


Figure 20: Rim cleaning and polishing process

The proposed automated system for (cleaning and) polishing car rims consists of several important components, each contributing to the overall efficiency and functionality of the polishing station. Below is a detailed breakdown of these components and their estimated costs:

Component	Estimated Cost (\$)
6 Dof Industrial Robot Arm	12,000-25,000
ATI Robotic Tool Changer	2,000-4,500
Polishing End-Effector	800-2,500
F/T Sensor	3,000-6,000
Fluid Delivery System	500-1,200
Cleaning Tool (Microfiber)	300-800
Pneumatic Actuator	300-700
ACF Control System	3,000-7,000

Table 18: Estimated Cost of Station

3.5 Organized Storage

The first design consideration is that the rims exit the production line unpackaged, requiring careful handling to prevent damage. Additionally, front and rear rims may vary in model and specifications, necessitating a storage system where each rim is placed in an individual bin to maintain organization and accessibility. Furthermore, order quantities are unpredictable, meaning the system must support both single-item and multiple-item retrieval while avoiding stacking. By ensuring each rim remains independent, the system allows seamless order fulfillment without unnecessary repositioning or manual effort.

The proposed storage system consists of a flat sliding rim entry system, a pre-positioned inclined storage bin, an AMR-based transport mechanism, and an automated vertical storage system. The flat sliding rim entry system ensures that rims transition smoothly from the conveyor belt into storage. Instead of allowing them to roll freely, the system keeps them in a stable, flat-laying position, reducing the risk of misalignment or damage. The inclined storage bins are designed to guide each rim into a precisely centered position, ensuring uniform placement while also protecting the rim's surface with soft silicone or foam padding. Once the rim is securely positioned in its bin, an Autonomous Mobile Robot (AMR) collects and transports the bin to either a storage area or a temporary buffer zone, as shown in Fig. 21. This method enables efficient and flexible handling of rims without the need for complex robotic arms or gripping mechanisms.

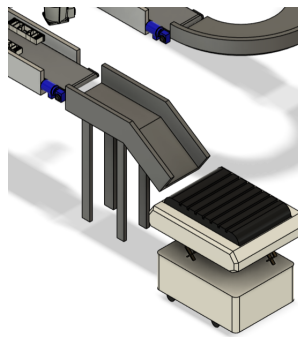


Figure 21: CAD Model of the Autonomous Mobile Robot for Storage

The vertical storage racks allow rims to be stored in an optimized high-density configura-

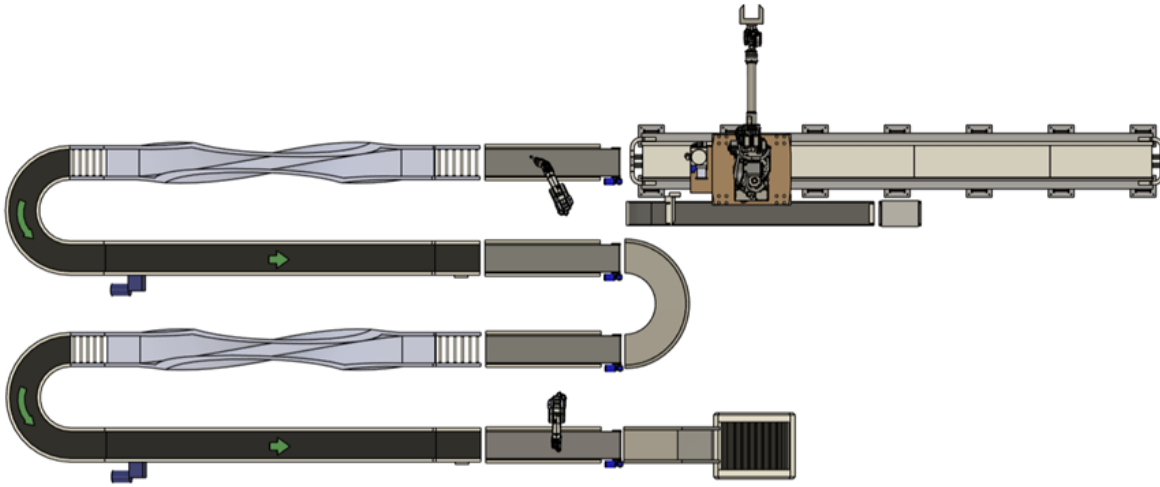


Figure 22: Whole Production Line

tion. Since each rim is stored individually, the system provides flexibility for both single-item and batch retrieval while maintaining efficient inventory tracking. Compared to alternative sunken storage designs, the vertical storage system offers a compact layout that can be expanded as needed. The modular nature of the system allows for easy scaling without major infrastructure changes. Furthermore, the separation between the production line and the storage area, facilitated by the AMR transport mechanism, ensures a smooth and automated workflow.

This design presents several key advantages. It fully supports both single-item storage and retrieval, making it suitable for dynamic order fulfillment. The system eliminates the need for complex robotic gripping solutions, reducing costs and technical challenges. Its modular and expandable design ensures high reliability and adaptability to future storage requirements. By utilizing a combination of mechanical guidance and gravity-assisted sliding, the system maintains operational stability and requires minimal maintenance.

4 Conclusion

The rim is picked up by a robotic arm and placed on a conveyor belt. The conveyor transports the rim to a fixed position, where it is fixed by a clamp to ensure stability during operation. The rim then undergoes deburring. After deburring, the rim passes through a gravity-based flipping mechanism and is circulated back within the working range of the robotic arm for another round of deburring. It is then conveyed to the polishing zone, where it goes through a similar process. Finally, the finished rim is transported and stored by an AMR

	0.5min	1min	0.5min	3min	0.5min	1.5min	0.5min	1min	0.5min	2min	0.5min	2min	0.5min	2min
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14
Drone Scanning	S1	-	-	-	-	-	S2	-	-	-	-	-	-	-
Rim Retrieval	-	P1	T1	-	-	-	-	-	-	-	-	-	-	-
Deburring	-	-	R1	D1A	FA1	D1B	D1B	D1B	T1	-	-	-	-	-
Polishing	-	-	-	-	-	-	-	-	R1	P1A	FB2	P1B	T1	-
Storage	-	-	-	-	-	-	-	-	-	-	-	-	R1	ST1

Figure 23: Non-Uniform Time Slot Pipeline Table

This is our production line's Non-Uniform Time slot Pipeline Table, the time slot is divided by events. S_n : Scan the nth rim P_n : Robotic gripper picks up the nth rim T_n : Conveyor sends away the nth rim R_n : Conveyor receives the nth rim D_nA/D_nB : Deburring robot processes Side A/B of the nth rim P_nA/P_nB : Polishing robot processes Side A/B of the nth rim ST_n : Automated storage system stores the nth rim

The results show that one complete iteration takes approximately 16 minutes.

By creating a Uniform Time slot Pipeline Table based on this workflow, it can be observed that after the production line becomes operational, this solution achieves a 9-minute time saving per cycle. Specifically, the drone will have 6.5 minutes of idle time between two consecutive cycles, the gripping robotic arm will experience 5.5 minutes of idle time, the deburring robotic arm operates continuously without any downtime, the polishing robotic arm has 1 minute of idle time, and the vertical storage system will have 8 minutes of idle time during the process.

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15
Drone Scanning	S1	-	-	-	-	-	-	-	-	-	-	-	-	-	S2
Rim Retrieval	-	P1	P1	T1	-	-	-	-	-	-	-	-	-	-	-
Deburring	-	-	-	R1	D1A	D1A	D1A	D1A	D1A	D1A	FA1	D1B	D1B	D1B	D1B
Polishing	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Storage	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

	T16	T17	T18	T19	T20	T21	T22	T23	T24	T25	T26	T27	T28	T29	T30
Drone Scanning	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Rim Retrieval	P2	P2	T2	-	-	-	-	-	-	-	-	-	-	-	-
Deburring	D1B	D1B	T1/R2	D2A	D2A	D2A	D2A	D2A	D2A	FA2	D2B	D2B	D2B	D2B	D2B
Polishing	-	-	R1	P1A	P1A	P1A	P1A	FB1	P1B	P1B	P1B	P1B	T1	-	-
Storage	-	-	-	-	-	-	-	-	-	-	-	-	R1	ST1	ST1

	T31	T32	T33	T34	T35	T36	T37	T38	T39	T40	T41	T42	T43	T44	T45
Drone Scanning	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Rim Retrieval	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deburring	D2B	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Polishing	R2	P2A	P2A	P2A	P2A	FB2	P2B	P2B	P2B	P2B	T2	-	-	-	-
Storage	ST1	ST1	-	-	-	-	-	-	-	-	R2	ST2	ST2	ST2	ST2

T=0.5min

Figure 24: Uniform Time Slot Pipeline Table - every time slot is same

While the current design demonstrates an efficient and scalable automation workflow, an important consideration remains, the number of robotic arms employed at each processing station. Introducing two robots per station—particularly in the deburring and polishing zones—would significantly reduce processing time and increase throughput, allowing more

rims to be treated within the same operational window. This dual-robot configuration would help address bottlenecks and ensure continuous operation even if one robot is engaged or requires maintenance. However, this improvement comes at the cost of increased capital investment, maintenance complexity, and space requirements. Conversely, limiting each station to a single robotic arm reduces overall system cost and simplifies integration, but may lead to longer cycle times, especially when handling rims of varying sizes and surface conditions. Therefore, the decision must balance processing speed, cost constraints, and the expected scale of production, introducing a trade-off between processing time and cost. With the current single-robot design offering a cost-effective yet expandable foundation.

4.1 Future Improvements

Overall, the proposed automated storage system for rims successfully integrates gravity-assisted sliding with AMR-based transportation, ensuring efficient and modular handling. By employing a structured yet simple design, the system eliminates the need for robotic arms and complex vision-based gripping solutions, thereby reducing costs and maintenance complexity. The use of pre-positioned tilted bins ensures precise rim placement while preventing surface damage, and the AMR-based transport system seamlessly connects the production line to the storage facility.

However, the current design has some limitations, particularly in terms of space utilization. The flipping slope occupies a significant footprint, which may hinder future expansions. To address this, future improvements could include a recycling flipping slope, where rims are transported back to the same flipping mechanism using a conveyor belt, minimizing redundant structures. Another alternative is a spiral flipping slope, which compresses the flipping mechanism into a vertical format, reducing horizontal space requirements.

These optimizations would improve space efficiency and scalability, making the system more adaptable to future smart warehouse implementations.

4.2 Cost Considerations

The proposed automation system involves significant initial investments, but the long-term benefits in efficiency, precision, and labor savings justify the expenditure. A breakdown of estimated costs for key components includes:

1. Rim Retrieval System

Total Retrieval System Cost: \$33,000–62,500

2. Deburring Station

Total Deburring Station Cost: \$20,300–44,700

3. Polishing Station

Total Polishing Station Cost: \$22,900–49,700

4. Flipping Mechanism & Conveyor

Total Flipping Mechanism Cost: \$5,000–10,000

Component	Estimated Cost (USD)	Products
4× Drones with LiDAR	\$10,000–16,000	Comparable to DJI Enterprise drones with LiDAR (DJI Matrice 300 with L1 Lidar)
Embedded CNN Processor	\$800–1,500	NVIDIA Jetson AGX Orin or similar edge AI modules (NVIDIA Jetson Pricing)
6-DoF Robotic Arm	\$12,000–25,000	UR10e cobot or industrial arms like Fanuc LR-10iA (Universal Robots Pricing)
3-Finger Adaptive Gripper	\$2,000–4,000	OnRobot 3FG15 or Robotiq Hand-E (OnRobot Grippers)
Motorized Linear Axis	\$5,000–10,000	Bosch Rexroth or Hiwin linear rails (Bosch Rexroth Catalog)

Table 19: Cost breakdown for rim retrieval system

Component	Estimated Cost (USD)	Reference/Justification
6-DoF Industrial Robot Arm	\$12,000–25,000	UR10e cobot or industrial arms like Fanuc LR-10iA (Universal Robots Pricing)
Force-Torque Sensor	\$3,000–6,000	ATI Axia80 or OnRobot HEX (ATI Industrial Automation)
Deburring End-Effector	\$1,000–3,000	Custom tooling or off-the-shelf abrasive systems (3M Deburring Tools)
Debris Extraction System	\$800–2,000	Industrial vacuum systems (Festool Dust Extractors)

Table 20: Cost breakdown for deburring station

5. Organized Storage (AMR + Bins)

Total Storage System Cost: \$15,000–30,000

Total Estimated System Cost is around 96,200–196,900 USD

While the upfront cost is high, automation reduces manual labor, minimizes errors, and increases throughput, leading to a strong return on investment over time.

Component	Estimated Cost (USD)	Reference/Justification
6-DoF Robot Arm	\$12,000–25,000	UR10e cobot or industrial arms like Fanuc LR-10iA (Universal Robots Pricing)
Force-Torque Sensor	\$3,000–6,000	ATI Axia80 or OnRobot HEX (ATI Industrial Automation)
Polishing End-Effector	\$800–2,500	Rotary tools with microfiber pads (FLEX Polishers)
Pneumatic Actuator	\$300–700	Festo or SMC linear actuators (Festo Catalog)

Table 21: Cost breakdown for polishing station

Component	Estimated Cost (USD)	Reference/Justification
PU-Coated Conveyor Belt	\$2,000–5,000	Dorner or Intralox belts (Dorner Conveyors)
Photoelectric Sensors	\$500–1,500	Keyence or Omron sensors (Keyence Catalog)
Structural Framework	\$2,500–3,500	Custom fabrication (market rates for steel/aluminum)

Table 22: Cost breakdown for flipping mechanism

Component	Estimated Cost (USD)	Reference/Justification
Autonomous Mobile Robot (AMR)	\$10,000–20,000	MiR100 or Fetch Core (Mobile Industrial Robots)
Vertical Storage Racks	\$5,000–10,000	Modular shelving systems (Interlake Mecalux)

Table 23: Cost breakdown for organized storage

References

- [1] ATI Industrial Automation, “Axia80-m20 force/torque sensor,” 2024. Accessed: 2024-03-31.
- [2] OnRobot, “Hex 6-axis force/torque sensor,” 2024. Accessed: 2024-03-31.
- [3] 3M Industrial Abrasives Finishing, “Scotch-brite™ surface conditioning wheels selection guide,” 2024. Accessed: 2024-03-31.
- [4] GRAM Conveyor, “Conveyor belt material: History applications,” 2024. Accessed: 2024-03-23.
- [5] HT Conveyor, “10 belt conveyor types & 5 types of conveyor belt materials & their applications,” 2024. Accessed: 2024-03-23.
- [6] M. Soori, “Manufacturing processes of car alloy wheels,” 2022.
- [7] ChipsAway, “How to polish alloy wheels,” 2024. Accessed: 2024-03-25.
- [8] 3M, “Perfect-it ex paint finishing brochure,” 2018. Accessed: 2024-03-25.
- [9] FLEX Tools, “Pe 8-4 80 rotary polisher,” 2024. Accessed: 2024-03-25.
- [10] ATI Industrial Automation, “Robotic tool changers,” 2024. Accessed: 2024-03-26.
- [11] OnRobot, “The power and importance of sensing technologies,” 2024. Accessed: 2024-03-25.
- [12] I. Mohsin, K. He, Z. Li, and R. Du, “Path planning under force control in robotic polishing of the complex curved surfaces,” *Applied Sciences*, vol. 9, no. 24, p. 5489, 2019.